

KARIM ALMOMANI

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EDUCATION

McMaster University

Hamilton, Canada

B.Eng. in Computer Engineering (Co-op program), 3rd Year, GPA: 3.70/4.00

Program Duration: 2024 – 2029

- Related Coursework: Microprocessor Systems Project, Digital Systems Design, Electronic Devices and Circuits II, Electronic Devices and Circuits 1, Logic Design, Introduction to Circuit Analysis, Data Structures and Algorithms.

EXPERIENCE

EcoCAR McMaster – 2nd place EV Design Competition Team

Oct 2025 – Present

Systems Design & Integration - High Voltage Team Member

Hamilton, Canada

- Maintained and reinstalled **automotive inverters** and high voltage cables, while documenting high-voltage safety protocols complying with EcoCAR competition standards.
- Co-presented** a hands-on PCB design workshop in **Altium Designer**, training peers on schematic layouts and PCB routing.
- Performed soldering tasks on PCBs for the Vehicle Inspection subteam, validating custom test hardware connections.

Orange Telecommunications

May 2025 – Aug 2025

Software Development Intern

Amman, Jordan

- Developed responsive web pages integrating **APIs**, HTML5, CSS, Bootstrap, and **JavaScript**, delivering user-friendly UIs.
- Mentored incoming interns by setting onboarding goals and providing ongoing technical feedback, building a collaborative environment.

PROJECTS

3D Environment Scanner | Microprocessor Systems Project

Winter 2026

- Engineered **Embedded C firmware** on an **ARM Cortex-M4** to rotate a Time-of-Flight sensor via **I2C** and a GPIO-controlled stepper motor, taking multiple 360° scans and joining them to reconstruct a live 3D model.
- Implemented GPIO polling for sensing button inputs (scan/pause/return), & **UART** sending CSV data to the PC
- Built a **Python** program using **PySerial** to read the serial port, parse data and flag anomalies, then creating a 3D render of the surroundings using **Matplotlib**
- Debugged UART communication using **RealTerm**, firmware execution with **Keil Debugger**, and Python visualization using test datasets, resulting in a robust device.

Radial Object Detection System | Personal Project

Summer 2026

- Engineered a **real-time** object detection system, integrating a stepper motor and ToF sensor using **FreeRTOS** to detect the presence of solid objects within a **3-meter radius**.
- Utilized **FreeRTOS** to manage tasks, interrupts, and UART streaming for proximity data, on an ARM Cortex-M4, achieving **<40ms sensor reaction** latency

3-Bit Digital-To-Analogue Converter | Electronic Devices and Circuits

Winter 2026

- Designed and simulated a 3-bit binary-weighted-resistor DAC using **LTSpice** with an inverting summing op-amp, achieving **0.714 V/step** over 0-5V, while limiting input current draw to **<10nA** by utilizing buffer and comparator op-amps.
- Designed the finished circuit on **Altium Designer**, laying out a 2-layered PCB
- Tested performance using **Digilent Analog Discovery 2's oscilloscope**, measuring **-1% gain** error

Digital Logic Design Labs | Logic Design

Fall 2025

- Simulated sequential circuits by implementing finite state machines on **FPGA** boards using **Intel Quartus**

EV Charger Finder | Personal Project

Summer 2025

- Developed a responsive web application using **JavaScript** and BootStrap, to filter and locate compatible EV charging stations within a specified region
- Integrated third party APIs and the leaflet.js map library to display charger locations on a user-interactive map

Snake Game | C++ Course Project

Fall 2025

- Developed an **Object Oriented** CLI Snake game in **C++** using pointers and scalable dynamic memory allocation
- Implemented core game logic, managing input handling and memory management

TECHNICAL SKILLS

Programming: C, SystemVerilog, C++, Python, JavaScript

Embedded Systems: Embedded C, ARM Cortex-M, FreeRTOS, UART, I2C, FPGA, PySerial

Tools: Keil uVision, Altium Designer, Intel Quartus, LTSpice, Digilent WaveForms, RealTerm, Git, GitHub, Autodesk Inventor (CAD)